

RoadSense AI — Decisions To Make

These are the choices **only you can make** before the build work can proceed. Everything else in `PRD_Review_and_Actions.md` (PRD edits, code, tests) waits on these.

Each decision below: **the question · the options · my recommendation · what it unblocks**. Date: 2026-07-18

✓ ALL 7 DECIDED (2026-07-18)

#	Decision	Call made
1	Cloud vs edge	Cloud now, edge later
2	Class scope	4 classes + water flag (pothole, crack, broken road, edge damage; drop speed breaker + patch repair)
3	Weather field	Keep, label "for future use"
4	#1 user	Authority first ⚠️ needs a data-seeding plan — authorities need a filled map to see value
5	Real dashcam footage	Yes — commit (highest-value action)
6	Privacy (blur faces/plates)	Commit, build before launch (not blocking private testing)
7	Accuracy target	Measure first, then commit (let the footage set the number)

Details and reasoning for each below.

Decision 1 — Accuracy target (the D-4 blocker) ● highest priority

Question: The PRD demands **precision ≥95% / recall ≥90%**. This is *proven unreachable* by more data or more training (the ceiling is label quality). What is the new, honest target?

Option	Meaning
A. Keep 95%	Dishonest — already falsified twice. Not recommended.
B. Re-baseline on real footage (recommended)	Target = "detects X% of potholes on real dashcam footage, <5% false positives." Measured, defensible.
C. Invest in relabelling	The only path with real headroom, but slow + expensive. A separate project.

➡️ **Recommendation: B** — set the target from the real-footage ship gate you already use. Requires Decision 5 (get footage) to set the exact number. **Unblocks:** Non-functional Requirements, Success Metrics, the whole "is it done?" question.

Decision 2 — Who is the #1 user?

Question: Authorities and Fleets want different things. Who do you design and sell for *first*?

Option	Meaning
A. Authority	The buyer who fixes roads. Dashboard optimised for repair decisions.
B. Fleet	The data source. Product optimised for easy camera deployment + vehicle protection.

→ **Recommendation:** lean Fleet as the data engine, Authority as the buyer — but you must pick the *primary* one, because it decides who the dashboard is built for and who pays. **Unblocks:** Target Users, dashboard priorities, business model.

Decision 3 — Cloud now, edge later?

Question: Does the AI run in the cloud (built) or on a chip in the vehicle (edge, not built)?

Option	Meaning
A. Cloud now, edge later (<i>recommended</i>)	Ship what's built; move to edge at scale when bandwidth forces it.
B. Edge now	Months of new work (hardware + shrinking the model). Not ready.

→ **Recommendation:** A — already validated in our discussion; just needs to be made official + written into the PRD. **Unblocks:** System Overview, Hardware (smartphone = beta device), frame-sampling work.

Decision 4 — Class scope: how many?

Question: The PRD lists 7 detection classes. Each new class needs thousands of hand-labels. How many?

Option	Meaning
A. 4 classes + water flag (<i>recommended</i>)	Pothole, Crack, Broken Road, Road Edge Damage. Drop Speed Breaker + Patch Repair (wrong for this product); Water-filled = a flag.
B. All 7	Expensive, and 2 don't fit a road-damage product.

→ **Recommendation:** A — build 4 that matter; drop the 2 that aren't road damage. **Unblocks:** Detection Classes, AI Pipeline, the labelling roadmap.

Decision 5 — Get real dashcam footage? (the practical unlock) ●

Question: Every accuracy claim is from **handheld** footage. No model has been tested from a moving vehicle. Will you commit to getting real dashcam footage + one recorded drive?

Option	Meaning
A. Yes — get footage (<i>strongly recommended</i>)	Unblocks Decision 1's number, GPS validation, and the biggest untested risk — all at once.
B. No	Then the product stays unproven and Decision 1 can't be set honestly.

➔ **Recommendation: A** — this is the single highest-value action in the entire review. **Unblocks:** Decision 1, GPS tests, field validation (A3-5), real-device latency.

Decision 6 — Weather field: keep or drop?

Question: Weather is captured but shown nowhere. Keep it or remove it?

Option	Meaning
A. Keep, labelled "future use" (<i>recommended</i>)	Cheap to store; may feed degradation prediction later. Just don't call it a feature.
B. Surface it now	Only if a dashboard actually uses it.
C. Drop the column	If it will never be used.

➔ **Recommendation: A** — keep collecting, mark "for future use." **Unblocks:** Data Captured, Dashboards.

Decision 7 — Privacy: commit to blurring faces & plates?

Question: Road photos capture pedestrians' faces and licence plates = personal data, a legal issue for real deployment. Commit to blurring before storage?

Option	Meaning
A. Yes — add blurring (<i>recommended before any public deployment</i>)	Required to be lawful. Adds a processing step.
B. Defer	Acceptable only for closed testing, never for a real launch.

➔ **Recommendation: A** for anything public; B is fine only while testing privately. **Unblocks:** Security, deployment readiness.

Summary — decide these, in this order

#	Decision	Priority
5	Get real dashcam footage	● Do first — unblocks the most
1	Re-baseline accuracy target	● Blocks "is it done?" (needs #5)
2	#1 user (Authority vs Fleet)	● Shapes product + business
3	Cloud now / edge later	● Confirm + document
4	Class scope 7 → 4	● Confirm + document
6	Weather keep/drop	● Quick call
7	Privacy blurring	● Before any public launch

Once these 7 are decided, the PRD edits, code work, and tests in the main doc can all proceed.
